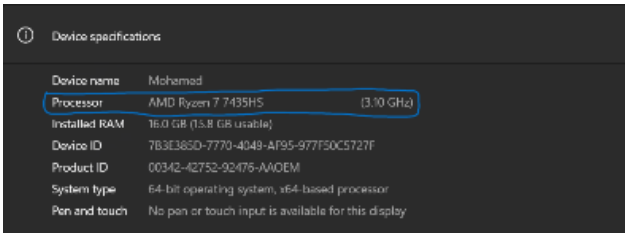
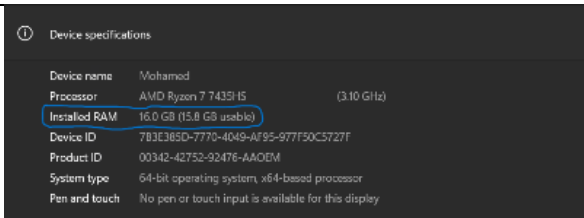
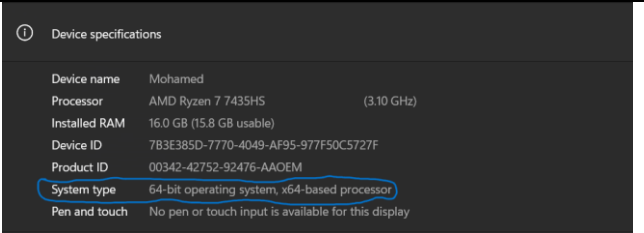
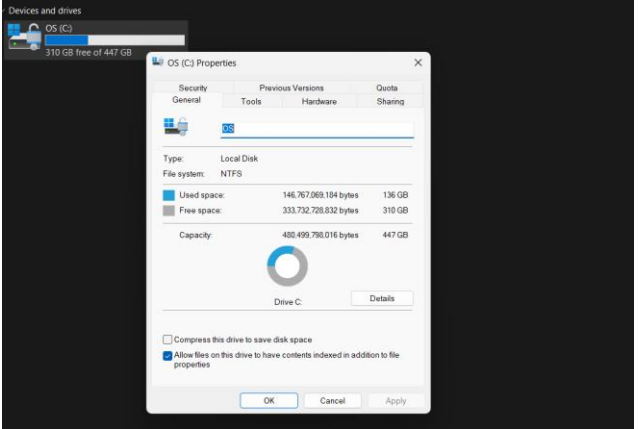
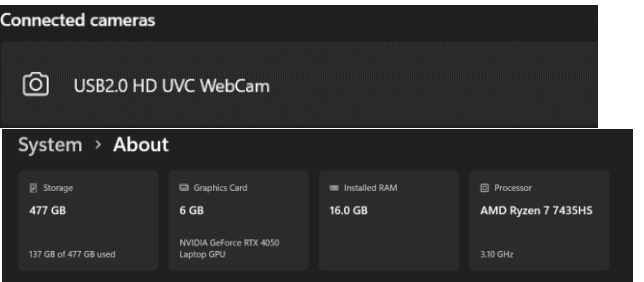
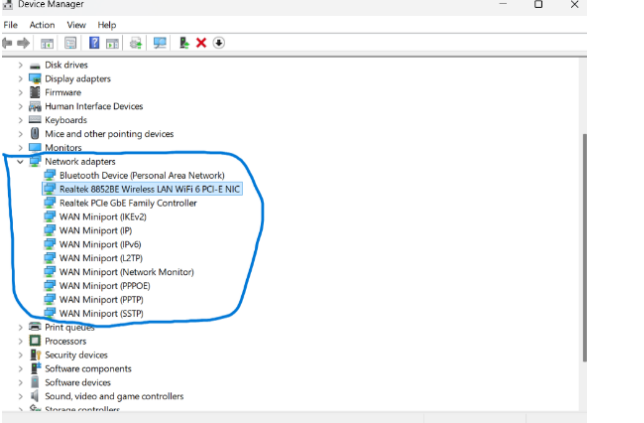
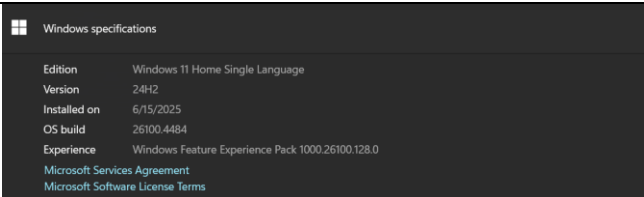


**Computer Architecture**  
**Assignment 1 (38m; 10%)**  
**(ANSWER ALL QUESTIONS)**  
**CLO2-PLO2-C2**

1. List the specification of your computer for the following hardware components with pictures (10m):
- Computer type
  - Computer brand/model
  - Processor
  - RAM (size)
  - System Type
  - Hard disk
  - Multimedia components
  - Network components
  - Operating System.

| No. | Hardware Components  | Specifications                        | Picture                                                                              |
|-----|----------------------|---------------------------------------|--------------------------------------------------------------------------------------|
| 1.  | Computer type        | Laptop                                |   |
| 2   | Computer brand/model | ASUS TUF Gaming A15 FA507NUR_FA507NUR |  |
| 3   | Processor            | AMD Ryzen 7 7435HS (3.10 GHz)         |  |
| 4   | RAM (size)           | 16.0 GB (15.8 GB usable)              |  |

|   |                       |                                                                                                    |                                                                                      |
|---|-----------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 5 | System Type           | 64-bit operating system, x64-based processor                                                       |    |
| 6 | Hard disk             | 477 GB SSD (136 GB used)                                                                           |    |
| 7 | Multimedia components | NVIDIA GeForce RTX 4050 Laptop GPU (6 GB), Built-in speakers, microphone, and USB2.0 HD UVC webcam |   |
| 8 | Network components    | Wi-Fi 6, Bluetooth                                                                                 |  |
| 9 | Operating System.     | Edition Windows 11                                                                                 |  |

2. Find brochures and compare two models of home computers for sale. Compare the specification and the prices of the computers.  
Choose which computer you want to buy, and state the reasons of your choice based on:
  - Price
  - Specification/performance
  - Purpose of use

| Specification     | Model 1: HP Pavilion Desktop PC TP01         | Model 2: Acer Aspire TC-1780                   |
|-------------------|----------------------------------------------|------------------------------------------------|
| Processor (CPU)   | Intel® Core™ i5-13400                        | Intel® Core™ i5-13400                          |
| Graphics (GPU)    | NVIDIA® GeForce® RTX 3050 (8 GB)             | Intel® UHD Graphics 730 (Integrated)           |
| Memory (RAM)      | 16 GB DDR4 RAM                               | 8 GB DDR4 RAM                                  |
| Storage           | 1 TB NVMe SSD                                | 512 GB NVMe SSD                                |
| Included          | CPU Tower, HP 23.8" Monitor, Keyboard, Mouse | CPU Tower, Acer 23.8" Monitor, Keyboard, Mouse |
| Connectivity      | Wi-Fi 6, Bluetooth 5.3                       | Wi-Fi 6, Bluetooth 5.2                         |
| Price (Estimated) | RM 4,199                                     | RM 2,899                                       |

**My Choice:** The **HP Pavilion Desktop PC TP01**.

**Reasons:**

- **Price:** While the HP costs more, it offers much better value. The extra RM 1,300 gets you upgrades that are worth far more than the price difference.
- **Specification/Performance:** The HP is significantly more powerful.
  - **Graphics:** Its NVIDIA® RTX 3050 graphics card is vastly superior for gaming and creative tasks compared to Acer's basic integrated graphics.
  - **Memory & Storage:** It has double the RAM (16 GB) for better multitasking and double the storage (1 TB) for your files and applications.
- **Purpose of Use:** The HP is the better all-around computer. It can handle demanding tasks like modern gaming, video editing, and design work. The Acer is only suitable for basic web browsing and office documents. The HP will perform better and last longer.

3. List FIVE types of computer and state the following for each type:

- Purpose of use
- Current popular brand
- Picture
- Price

| No | Type       | Purpose                                                 | Brand | Picture                                                                              | Price      |
|----|------------|---------------------------------------------------------|-------|--------------------------------------------------------------------------------------|------------|
| 1  | Laptop     | To do schoolwork, gaming, and browse the internet.      | ASUS  |    | RM4199     |
| 2  | Smartphone | To communicate via calls and social media apps.         | Apple |   | RM4299     |
| 3  | Tablet     | To watch movies and read digital books.                 | Honor |  | RM1800     |
| 4  | Server     | To host a company's website.                            | Dell  |  | RM6000     |
| 5  | Mainframe  | Processes huge volumes of transactions (e.g., banking). | IBM   |  | RM12694500 |

4. Make a timeline for the following (9m):

- a. Computer from 1940 - 1979
- b. Computer processor since 1980.
- c. Personal Computer since 1980.

a. 1940–1979: Computers

- 1943–1944 – Colossus (UK), used for codebreaking
- 1946 – ENIAC, first general-purpose electronic computer
- 1951 – UNIVAC I, first commercial computer in the US
- 1956 – IBM 305 RAMAC, first computer with a hard disk drive
- 1964 – IBM System/360, family of compatible mainframe computers
- 1971 – Intel 4004, first microprocessor
- 1973 – Xerox Alto, early personal workstation with GUI
- 1975 – Altair 8800, first popular microcomputer
- 1977 – Apple II, first mass-market personal computer
- 1979 – VisiCalc, first spreadsheet software for PCs

b. 1980–Present: Computer Processors

- 1981 – Intel 8088 used in IBM PC
- 1985 – Intel 80386 introduced 32-bit processing
- 1993 – Intel Pentium released
- 2000 – Intel Pentium 4 and AMD Athlon
- 2006 – Intel Core series launched
- 2011 – Intel Sandy Bridge and AMD Bulldozer
- 2017 – AMD Ryzen and Intel Core 8th Gen
- 2020 – Apple M1 ARM-based chip
- 2023 – Intel 13th Gen, AMD Ryzen 7000 series

c. 1980–Present: Personal Computers

- 1981 – IBM PC released
- 1984 – Apple Macintosh introduced GUI
- 1990 – Microsoft Windows 3.0 popularized GUI on PCs
- 1995 – Windows 95 released, plug and play support
- 2001 – Windows XP and Mac OS X launched
- 2007 – Rise of ultrabooks and netbooks
- 2010 – iPad started the tablet computing trend
- 2020 – Shift to ARM-based personal computers (Apple M1)
- 2024 – AI PCs with neural processing units (NPUs) emerging